

**a****Published methods for Ketobemidone**

4 independent literature protocols, one pollutant

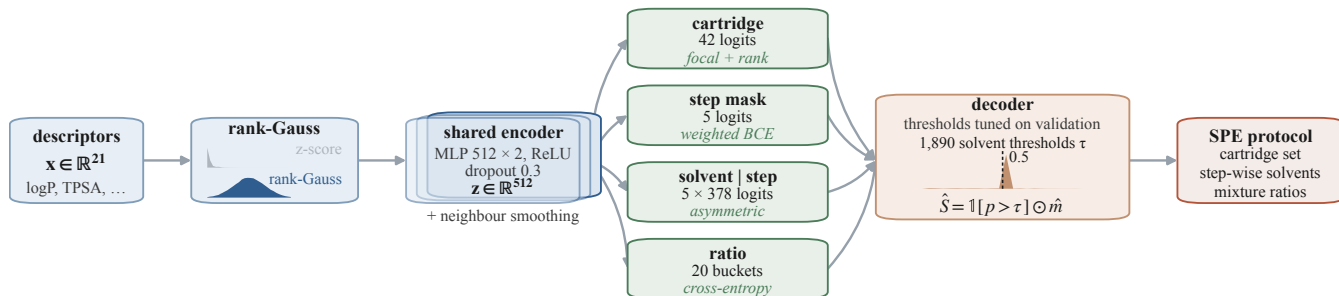
|          | Load                     | Condition                             | Wash                                              | Elute                                    | Reconstitute                           |
|----------|--------------------------|---------------------------------------|---------------------------------------------------|------------------------------------------|----------------------------------------|
| method 1 | MeOH+phosphate           | MeOH<br>H <sub>2</sub> O<br>phosphate | phosphate<br>MeOH                                 | EtOAc+hexane<br>NH <sub>4</sub> OH+EtOAc | MeCN+HCOOH                             |
| method 2 | K-phosphate              | —                                     | —                                                 | —                                        | NH <sub>4</sub> HCO <sub>2</sub> +MeOH |
| method 3 | NH <sub>4</sub> OAc+MeOH | MeOH<br>H <sub>2</sub> O              | MeOH+H <sub>2</sub> O                             | MeCN                                     | NH <sub>4</sub> OAc+MeOH               |
| method 4 | HCOOH+H <sub>2</sub> O   | MeOH<br>H <sub>2</sub> O              | H <sub>2</sub> O<br>MeOH+H <sub>2</sub> O<br>MeOH | MeCN+H <sub>2</sub> O                    | MeOH+H <sub>2</sub> O                  |

▼ union over methods; repeated ratios averaged

**target**

|                                                                                     |                                       |                                                                |                                                                           |                                                                                                           |
|-------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| MeOH+phosphate<br>K-phosphate<br>NH <sub>4</sub> OAc+MeOH<br>HCOOH+H <sub>2</sub> O | MeOH<br>H <sub>2</sub> O<br>phosphate | phosphate<br>MeOH<br>MeOH+H <sub>2</sub> O<br>H <sub>2</sub> O | EtOAc+hexane<br>NH <sub>4</sub> OH+EtOAc<br>MeCN<br>MeCN+H <sub>2</sub> O | MeCN+HCOOH<br>NH <sub>4</sub> HCO <sub>2</sub> +MeOH<br>NH <sub>4</sub> OAc+MeOH<br>MeOH+H <sub>2</sub> O |
|-------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|

Four laboratories elute the same compound four different ways;  
scoring against any single one would mark the other three wrong.

**c**

× 8 bootstrap members; head probabilities averaged before decoding

**b****Set-valued supervision**

11,441 pollutants · 10,335 with protocol annotation

|          |                                                            |                          |
|----------|------------------------------------------------------------|--------------------------|
| <b>c</b> | <b>cartridge set</b><br>3.34 labels per pollutant          | $\{0,1\}^{42}$           |
| <b>m</b> | <b>step mask</b><br>91% of steps present                   | $\{0,1\}^5$              |
| <b>S</b> | <b>solvent   step</b><br>8.25 distinct solvents            | $\{0,1\}^{5 \times 378}$ |
| <b>R</b> | <b>mixture ratio</b><br>39.7% of positive cells supervised | $[0,1]^{5 \times 378}$   |

